

Vedav Kerlepalli Sudhakargupta

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EDUCATION

- **Lamar University** Beaumont, Texas(**Aug 2023 - Present**)
M.S in Management and Information Systems;
◦ Relevant Courses:
Data Mining, Data Analysis, ERP-Overview, Configuration, E-commerce, Business Intelligence, Introduction to Programming Bus Solutions
- **Dayananda Sagar College of Engineering** Bengaluru, India (**Aug 2019 - Jun 2023**)
B.E in Computer Science & Engineering;
Achievement: Awarded Best Project (Overall) for my Final Year Project 8.78/10 C.G.P.A

TECHNICAL SKILLS

- Languages: Python, R, SQL,C,C++
- Version Control: Git, GitHub
- Microsoft Office Suite: Word, PowerPoint, Excel, Microsoft Dynamics 365, Power Automate
- Data Analytics: Tableau, Power BI
- Libraries/Frameworks: Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, Keras, OpenCV, PyTorch
- Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence, GANs, Large Language Models(LLMs), GPT, BERT, RoBERTa, RAG (Retrieval Augmented Generation)
- Mathematics for ML: Probability, Statistics
- Methodologies and Tools: Agile, Kanban, Jira, Confluence
- Cloud Tools: Amazon EC2, Amazon S3, Amazon Athena, AWS Cloud, Amazon Kinesis, Azure AI Services, Azure OpenAI Service,

PROFESSIONAL EXPERIENCE

- **BayGrape, California, USA** Aug 2023 – Sep 2024
Data Engineer Intern
 - **Developed Innovative Wheel Recognition Tool:** Researched and integrated **image recognition** technologies using **Machine Learning, Deep Learning,** and Computer Vision, leading to the creation of a **proof of concept (PoC)** and **minimum viable product (MVP)** that enhanced customer engagement and optimized wheel listings for the ChatGPT WP-GPT bot.
 - **Configured and Trained WP-GPT for BayGrape:** Tuned and configured an **OpenAI GPT** model, training it with fitment data scraped from 120+ web pages to improve accuracy and customer experience.
 - **Created Computer Vision BayGrape Replacement Script:** Developed a script enabling virtual wheel try-ons by replacing car wheels in images with user-selected options, enhancing the customer purchase experience.
- **Amara Raja Group, Hyderabad, India** Aug 2021 – Jun 2023
Data Analyst Intern
 - Analyzed competitors by market research and constructed wireframes for development.
 - Performed financial analysis to determine progress against strategic plan, budget, forecast O Dealt with fine - tuning stored procedures to improve performance using SQL Tuning advisor.
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- **CareTeamPvt. Ltd., Maharashtra, India** Sep 2021 - Feb 2022
Frontend Developer Intern
 - Developed frontend components to improve the user experience of the companies applications.
 - Used **Google Cloud's Vertex AI** and **OpenCV** to analyze object color and dimensions, reducing table positives by **30%** and improving accuracy by **20%**.
 - Implemented **MobileNetSSD models** for Better object detection, making the system more precise and re.

PROJECTS

- **Xalas – AI-Driven Financial Analysis Assistant:**
 - Utilizing **OpenAI's** Assistant API, Xalas performs **automated, real-time analysis** on earnings call data, **extracting metrics** like revenue growth and profit margins for major firms (e.g., Lincoln Financial, Fidelity). This **AI integration** delivers actionable **insights** in seconds.
 - Embedded as a Voiceflow-powered widget, Xalas provides a natural language interface with multi-threaded memory, enabling seamless, continuous analysis.
 - Hosted on Replit, Xalas's backend handles **thousands of queries** per second, leveraging OpenAI for instant, high-quality financial comparisons, and is built to integrate with extensive enterprise datasets for **scalable AI-driven insights**.
- **Smart Startup Analyzer, Prediction Model, and Analysis Tool for Venture Capitalists Using Machine Learning :**
 - Developed a **machine learning**-based model to aid venture capitalists in evaluating startups, accurately categorizing investments as Bad, Risky, Good, or Great.
 - Developed **Random Forest Ensemble** models to analyze startups and predict investment success, achieving 85% accuracy in identifying profitable deals.
 - Engineered the **Python** model and converted it into a deployable pickle file, integrating it into the backend for efficient decision-making.
 - Engineered feature pipelines leveraging 17 crunchbase **datasets** with 370,000 data points, implementing **SQL** queries and **Python** scripts to extract and transform key metrics.
- **GestureBased Game Control:**
 - Developed an immersive gesture control system using **Python** and **OpenCV** to enable interactive and customizable gaming experiences beyond traditional keyboard/mouse inputs.
 - Engineered a real-time vision pipeline processing 30+ FPS from a webcam, applying optimizations like Gaussian blurring and HSV thresholding for fast and accurate hand tracking.
 - Designed modular architecture supporting racing, and sidescrolling games by dividing the FOV(Field of View) into quadrants mapped to discrete game control like acceleration and steering.

CERTIFICATIONS

- **Google Cloud:** [Google Cloud Training](#)
- **Database Programming with SQL:** [DataBase SQL](#)

PUBLICATIONS

- **Fake URL Detection Uisng Machine Learning and Deep Learning:**
International Journal of Scientific Research Technology (IJSIRT), ISSN NO:2456-2165:IJSIRT22DEC1128

CAMPUS INVOLVEMENT / EXTRACURRICULARS

- **Graphic Designer for the Student Involvement Office at The Lamar University Beaumont** Jan 2024 - Dec 2024
Created design elements like logos, posters, and branding products for various student organizations and their events